Stance Detection

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***Abstract*—Stance detection in rumour discussions is challenging because the label distribution is highly imbalanced and useful evidence appears at different levels (contextual semantics vs. lexical cues). Existing methods often rely mainly on one side, which limits minority-class recognition. In this paper, we propose an uncertainty-aware residual fusion framework for RumourEval-2019 stance classification. The model combines a semantic branch (RoBERTa) and a lexical branch (TF-IDF with a pretrained MLP). We estimate semantic uncertainty using the entropy of an auxiliary gate head, and use it to adaptively rescale lexical features through residual gating before final fusion and classification. This design keeps a stable lexical path while increasing lexical contribution when semantic confidence is low. Experiments show that our model clearly outperforms strong baselines: compared with BERT fine-tuning, it improves accuracy from 0.7021 to 0.8489 and Macro-F1 from 0.4471 to 0.6216. It also slightly exceeds the SemEval-2019 winning Macro-F1 (0.6187). Further error analysis shows remaining errors are mainly minority-to-`comment` confusion, indicating future work should further improve decision boundaries between weak stance and generic commentary.**

***Index Terms*—Stance detection, rumour discussion, uncertainty-aware residual fusion framework, imbalanced distribution label**

I. Motivation

Recently, the rapid development of social media platforms such as X, WeChat, and TikTok has led to an explosive growth in user-generated content. The increasing availability of massive amounts of textual data has created a need to automatically analyze the opinions, emotions, and stances expressed. Stance detection has been an active topic in NLP. Stance detection aims to identify the author's attitude or stance towards a specific target, typically classified into categories such as support, opposition, or neutrality.

It has been used to analyze ideological debates [1], to support automated fact-checking [2], and to understand discussions around rumours in online platforms [3]. Stance detection has therefore been studied across different domains and textual settings, especially in social media environments where conflicting viewpoints are explicitly articulated.

Recent surveys summarize the methodological development of stance detection [4]. However, despite substantial progress, limited attention has been paid to systematically examining how different formulations of stance detection contribute to false content detection. Stance detection can either be treated as a fact checking mechanism in itself—by aggregating positions toward a claim—or as a component within a larger pipeline that leverages stance signals to predict veracity.

II. Background

A. Dataset

SDQC support classification. Example 1:

u1: We understand that there are two gunmen and up to a dozen hostages inside the cafe under siege at Sydney.. ISIS flags remain on display #7News **[support]**
- **u2:** @u1 not ISIS flags **[deny]**
- **u3:** @u1 sorry - how do you know its an ISIS flag? Can you actually confirm that? **[query]**
  - **u4:** @u3 no she cant cos its actually not **[deny]**
- **u5:** @u1 More on situation at Martin Place in Sydney, AU LINK **[comment]**
- **u6:** @u1 Have you actually confirmed its an ISIS flag or are you talking shit **[query]**

SDQC support classification. Example 2:

u1: These are not timid colours; soldiers back guarding Tomb of Unknown Soldier after today's shooting #StandforCanada PICTURE **[support]**
- **u2:** @u1 Apparently a hoax. Best to take Tweet down. [deny]
- **u3:** @u1 This photo was taken this morning, before the shooting. [deny]
- **u4:** @u1 I dont believe there are soldiers guarding this area right now. [deny]
  - **u5:** @u4 wondered as well. Ive reached out to someone who would know just to confirm that. Hopefully get response soon. [comment]
    - **u4:** @u5 ok, thanks. [comment]

Fig. 1. Examples of tree structure threads of the dataset, where the label associated with each tweet is the target of the SDQC support classification task.

Figure 1 illustrates this structure using an example topic.

The source text presents a rumour.For stance detection tasks, there is no need to verify whether the rumor is correct. Each source post initiates a discussion thread structured as a tree, where subsequent replies are connected through conversational links. Within these discussion threads, users may express different orientations toward the original rumour by supporting, denying, querying, or commenting on it. This is il lustrated in figure 1 with a Figure 2example

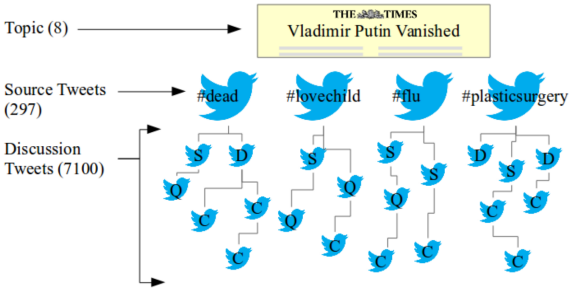

Fig. 2. Structure of the first rumours corpus

Formally, the class labels are defined as:

- **Support:** reply supports the rumour,
- **Deny:** reply denies the rumour,
- **Query:** reply asks for additional evidence,
- **Comment:** reply comments without explicit orientation.

It is important to note that the classification target is each individual reply in the discussion tree, rather than the source post itself. The task is therefore a target-dependent stance classification problem in a conversational setting.

RumourEval 2019 builds upon the RumourEval 2017 corpus and extends it in several directions. The 2017 corpus contains 297 source tweets grouped into eight topics, along with 7,100 associated discussion tweets. In the 2019 edition, this corpus serves as training data and is augmented with:

- New English Twitter test data,
- Reddit discussion data in English,
- Twitter data in Russian,
- Twitter data in Danish.

The dataset RumourEval 2019 consists of 7,005 training instances, 2,425 development instances, and 2,945 testing instances, each including a `source_text`, a `reply_text` and a corresponding stance label from the four way SDQC taxonomy.

Topics will be identified using Snopes and similar debunking projects. Potential source posts within these topics are identified on the basis of the amount of attention they attract; for tweets, number of retweets has been used success fully as an indicator of a good source tweet. Source texts are then manually selected from among these and labeled for veracity by an expert

Discussion threads are harvested using a previously established methodology. The stance annotations of discussion posts are obtained via crowdsourcing, with multiple annotators employed to ensure reliability. Quality control procedures and prior experience indicate high inter-annotator agreement for the SDQC labels.

Consistent with prior analyses on stance distributions, the dataset exhibits significant class imbalance, with *comment* being the most frequent class and *deny* and *query* occurring less often, which poses a challenge for balanced classification performance.

In summary, current stance detection methods still face two practical limitations on RumourEval-style data. First, due to strong class imbalance, minority stances (`support`/`deny`/`query`) are often confused with the dominant `comment` class. Second, existing models usually rely either on contextual semantic encoding or on lexical cues, but do not effectively balance these two sources at instance level. These observations motivate our method: an uncertainty-aware residual fusion framework that combines RoBERTa semantic representations with TF-IDF lexical representations, and dynamically adjusts lexical contribution according to semantic uncertainty.

B. Related work

1) Definition of Stance Detection: In order to understand the task of stance detection, we first clarify the notion of stance and the stance-taking process. Stance has been defined as the expression of a speaker's standpoint and judgement toward a given proposition [5]. We adopt the general formulation of stance detection proposed by Küçük and Can [4], which frames the task as a classification problem: given a textual input and a specified target, the goal is to predict the author's stance with respect to that target.

In some formulations, an additional *Neutral* category is included in the stance label set [6]. Furthermore, the target of evaluation may be explicitly mentioned in the text or must be inferred implicitly from context [6], [7].

2) Feature Engineering and Traditional Machine Learning Approaches: Early research on stance detection, particularly in the context of misinformation and fake news verification, relied heavily on manually engineered features combined with traditional machine learning classifiers. A prominent benchmark in this line of work is the Fake News Challenge (FNC), which evaluated systems using a weighted accuracy metric (FNC score) designed to mitigate the effects of class imbalance. Most participating systems employed hand-crafted features, including lexical features, word embeddings, sentiment lexica, and topic model representations [8], [9].

Subsequent analyses revealed that lexical features played the most significant role in system performance, while sentiment-based features contributed little to improvements [9]. Ghanem et al. [10] investigated the importance of lexical cues and found that report and negation features were particularly beneficial, whereas knowledge and denial cues were less effective.

However, these approaches exhibited notable limitations. In particular, models struggled to correctly identify minority classes such as *Disagree*, largely due to severe class imbalance [9]. Moreover, Hanselowski et al. [9] demonstrated that many systems implicitly relied on lexical overlap between the headline and the associated document, rather than modeling deeper semantic relationships. As a consequence, such models performed poorly in cases requiring semantic reasoning, complex negation handling, or broader propositional understanding. Reported results indicate that the *Disagree* class achieved very low F1 scores (around 18), while the *Unrelated* class was detected almost perfectly (over 99 F1) [9].

Overall, feature-based methods established important early baselines for stance detection but were constrained by their dependence on surface-level lexical signals and limited capacity for capturing deeper semantic structures.

3) Neural Approaches: Mohtarami et al. [11] explored memory network architectures to mitigate irrelevant information; however, performance on minority stance categories remained limited. This design also moves toward a more interpretable stance prediction process, as extracted snippets can serve as partial justifications for the predicted label.

With the advent of large scale pretrained language models, stance detection research shifted toward transfer learning frameworks based on Transformer architectures. Transformers has been explored [12], significantly improving the performance of previous state-of-the-art approaches. Guderlei and Aßenmacher [13] showed the most important hyper-parameter to be learning rate, while freezing layers did not help. In particular, using the pre-trained Transformer RoBERTa improved F1 from 18 to 58 for Disagree, and from 50 to 70 for Agree. The success of these models is also seen in cross-lingual settings. For Arabic, Khouja [14] achieved 76.7 F1 for stance detection on the ANS dataset using mBERT. Similarly, Hardalov et al. [15] applied pattern-exploiting training (PET) with sentiment pre-training in a cross-lingual setting showing sizeable improvements on 15 datasets. Alhindi et al. [16] showed that language-specific pre-training was pivotal, outperforming the state of the art on AraStance (52 F1) and Arabic FC (78 F1).

Notably, the use of RoBERTa substantially improved performance on minority classes, increasing F1 scores for *Disagree* and *Agree* categories compared to earlier neural models [13]. Beyond English, Transformer-based models have also proven effective in cross-lingual settings. For Arabic stance detection, Khouja [14] achieved strong performance using multilingual BERT (mBERT). Hardalov et al. [15] applied pattern-exploiting training (PET) in a cross-lingual setup, reporting consistent improvements across multiple datasets. Similarly, Alhindi et al. [16] demonstrated that language-specific pre-training significantly enhances performance on Arabic stance and fact-checking benchmarks.

III. Method

In this paper, we propose an uncertainty-aware residual fusion framework for stance classification, whicj explicitly combines two complementary views of each conversation instance: (i) a semantic view encoded by a contextual transformer, and (ii) a lexical view encoded by sparse n-gram statistics.

A key difficulty in multi-view fusion is that lexical cues can be highly informative for some instances, but also noisy or redundant when the semantic signal is already confident.

To address this, we estimate semantic uncertainty via the entropy of an auxiliary semantic head, and use it to modulate the contribution of lexical features through a residual gating mechanism.

An overview of the architecture is given in Fig. 3.

A. Overall Procedure

For clarity and reproducibility, the proposed method can be summarized in the following steps.

1. **Thread-aware input construction.** Apply a normalization function $\mathcal{N}(\cdot)$ that optionally replaces URLs and user mentions with placeholders (e.g., `$URL$`, `$MENTION$`). For raw conversation files, we compose the paired input as

$$(\tilde{r}_i, \tilde{s}_i) = \begin{cases} (\mathcal{N}(r_i), \mathcal{N}(p_i)), & \text{if source text is empty,} \\ (\mathcal{N}(r_i \oplus p_i), \mathcal{N}(s_i)), & \text{otherwise,} \end{cases} \tag{1}$$

where p_i is the parent reply and $\oplus$ denotes string concatenation. We also define the TF-IDF text as

$$t_i = \begin{cases} \tilde{r}_i, & \text{default,} \\ \tilde{r}_i \oplus \tilde{s}_i, & \text{optional variant.} \end{cases} \tag{2}$$

2. **Lexical encoding (TF-IDF).** Compute sparse TF-IDF vectors

$$v_i = \mathrm{TFIDF}(t_i) \in \mathbb{R}^V, \tag{3}$$

using unigram+bigram features, sublinear term frequency scaling, and vocabulary cap $V \leq 50{,}000$.

3. **Lexical pretraining (MLP).** Pretrain a two-layer MLP on top of TF-IDF:

$$h_i^t = \tanh(W_1 v_i + b_1), \quad h_i^t \in \mathbb{R}^{d_t}, \tag{4}$$

$$z_i^t = W_2 h_i^t + b_2, \quad z_i^t \in \mathbb{R}^C. \tag{5}$$

In our implementation, $d_t = 128$. The checkpoint is selected by validation macro-F1. In the subsequent fusion stage, the lexical branch parameters ϕ are frozen and only the representation h_i^t is used.

4. **Semantic encoding (RoBERTa).** Encode the paired input $(\tilde{r}_i, \tilde{s}_i)$ with RoBERTa:

$$H_i = f_\theta(\tilde{r}_i, \tilde{s}_i) \in \mathbb{R}^{L \times d_r}, \tag{6}$$

and use the first-token representation as sentence-level semantics:

$$h_i^r = H_i[0] \in \mathbb{R}^{d_r}. \tag{7}$$

For `roberta-base`, $d_r = 768$.

5. **Uncertainty estimation and residual gating.** Project semantic features to auxiliary logits

$$z_i^g = W_g h_i^r + b_g, \tag{8}$$

and compute a temperature-scaled distribution

$$p_i = \mathrm{softmax}(z_i^g / \tau). \tag{9}$$

The normalized entropy is

$$\bar{H}_i = \frac{-\sum_{c=1}^{C} p_{i,c} \log p_{i,c}}{\log C} \in [0, 1]. \tag{10}$$

We then obtain a scalar gate

$$\alpha_i = \sigma\big(s(\bar{H}_i - \beta)\big), \tag{11}$$

where s and β are learnable scalars. Importantly, the entropy in Eq. (10) is computed from the auxiliary gate logits z_i^g (from $W_g h_i^r + b_g$), which are used only for uncertainty estimation rather than final stance prediction.

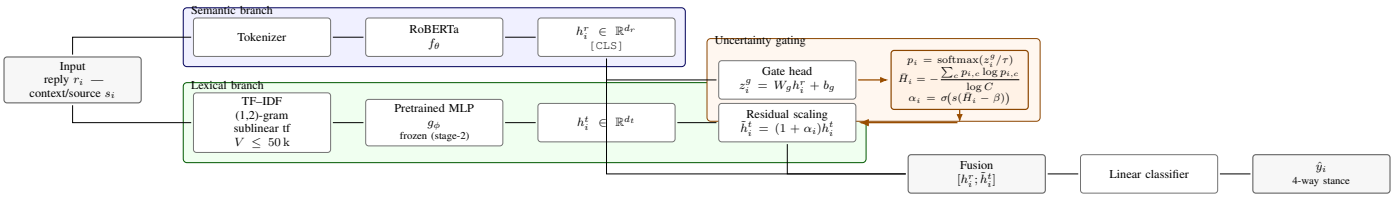

Fig. 3. Uncertainty-aware residual fusion. A semantic branch (RoBERTa) and a lexical branch (TF-IDF→MLP) are fused by entropy-guided residual scaling of lexical features.

The main variant performs residual scaling of lexical features:

$$\tilde{h}_i^t = (1 + \alpha_i)\, h_i^t. \tag{12}$$

This preserves an identity path for lexical information while allowing uncertainty-driven amplification.

6. **Fusion and prediction.** Concatenate semantic and gated lexical features:

$$h_i = [h_i^r; \tilde{h}_i^t] \in \mathbb{R}^{d_r + d_t}, \tag{13}$$

$$z_i = W_c h_i + b_c, \qquad \hat{p}_i = \mathrm{softmax}(z_i), \tag{14}$$

and predict $\hat{y}_i = \arg\max_c \hat{p}_{i,c}$.

B. Ablation Variants

To isolate the contribution of uncertainty-aware gating, we consider:

$$\tilde{h}_i^t = \begin{cases} h_i^t, & \texttt{concat baseline}, \\ \alpha_i h_i^t, & \texttt{multiplicative gate}. \end{cases} \tag{15}$$

C. Training Objective and Optimization

We optimize weighted cross-entropy:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} w_{y_i} \log \hat{p}_{i,y_i}. \tag{16}$$

Class weights are computed from training counts n_c as

$$w_c = C \cdot \frac{\left(\max(n_c, 1)^{-1}\right)^{\gamma}}{\sum_{j=1}^{C} \left(\max(n_j, 1)^{-1}\right)^{\gamma}}, \tag{17}$$

where γ controls the reweighting strength ($\gamma = 0$ recovers standard cross-entropy).

a) Two-stage training protocol.: (1) Pretrain the TF-IDF MLP with SGD and keep the checkpoint with best validation macro-F1. (2) Initialize the fusion model, freeze the pretrained lexical MLP, and train RoBERTa parameters θ together with the gate head and fusion classifier using AdamW.

D. Implementation Details

Unless otherwise stated, we use `roberta-base`, maximum sequence length 192–256, and dynamic padding. The lexical representation dimension is fixed to $d_t = 128$. A typical configuration is MLP pretraining for 55 epochs (learning rate 0.02, batch size 64), followed by fusion training for 6 epochs (learning rate 2×10^{-6}, train batch size 4, eval batch size 8). Our best entropy-gated setting uses residual gating (Eq. 12) with $\tau = 1.2$ and class-weight power $\gamma = 0.55$.

E. Design Rationale

The proposed fusion mechanism relies on the following intuition. RoBERTa captures contextual semantics and long-range dependencies, whereas TF-IDF+MLP captures sparse lexical cues that are often stance-indicative. By tying lexical amplification to semantic uncertainty (via entropy), the model increases the lexical contribution when the semantic branch is less decisive, while avoiding unnecessary reliance on lexical noise when semantic confidence is high. Finally, residual scaling preserves a stable lexical path and empirically yields robust optimization under class imbalance.

IV. Simulation Results

A. Evaluation metrics

To assess model performance, we report Accuracy, Macro-F1 and Weighted-F1. These metrics capture complementary aspects of classification quality and are particularly informative under class imbalance, which is common in stance detection (e.g., the `comment` class often dominates). Let K denote the number of classes. For each class $k \in \{1, \ldots, K\}$, we define TP_k (true positives), FP_k (false positives), FN_k (false negatives), and the class support n_k (number of ground-truth samples in class k). The total number of samples is $N = \sum_{k=1}^{K} n_k$.

The accuracy is defined as the proportion of correctly classified samples and is given by:

$$\mathrm{Accuracy} = \frac{1}{N} \sum_{k=1}^{K} \mathrm{TP}_k. \tag{18}$$

While easy to interpret, accuracy can be overly optimistic when the class distribution is highly skewed, as a model may achieve a high score by favouring the majority class while performing poorly on minority classes.

To better reflect class-wise performance, we compute precision, recall and F1-score for each class. The precision and recall of class k are given by:

$$\mathrm{Precision}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k}, \quad \mathrm{Recall}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FN}_k}. \tag{19}$$

The corresponding per-class F1 score, defined as the harmonic mean of precision and recall, is given by:

$$\mathrm{F1}_k = \frac{2\,\mathrm{Precision}_k\,\mathrm{Recall}_k}{\mathrm{Precision}_k + \mathrm{Recall}_k}. \tag{20}$$

Macro-F1 averages the per-class F1 scores by assigning equal weight to each class and is given by:

$$\text{Macro-F1} = \frac{1}{K}\sum_{k=1}^{K} \mathrm{F1}_k. \tag{21}$$

This metric treats all classes as equally important and is therefore sensitive to minority-class performance, making it suitable for verifying whether the model recognises minority stance categories.

Weighted-F1 averages the per-class F1 scores weighted by class support and is given by:

$$\text{Weighted-F1} = \sum_{k=1}^{K} \frac{n_k}{N}\,\mathrm{F1}_k. \tag{22}$$

Unlike Macro-F1, it reflects performance under the natural label distribution and is often closer to accuracy, while still accounting for the precision–recall trade-off.

Overall, using these three metrics jointly provides a comprehensive evaluation: accuracy summarizes global correctness, Macro-F1 emphasizes balanced performance across classes, and Weighted-F1 offers a distribution-aware view that better matches real-world frequencies.

B. Test results

Model	Acc	Macro-F1	W-F1
TF-IDF + Logistic Regression	0.6436	0.2500	0.6789
Sentence Embeddings + Logistic Regression	0.6507	0.2562	0.6781
BERT + Fine-tuning	0.7021	0.4471	0.7300
BranchLSTM	0.841	0.493	–
BLCU-NLP (SemEval-2019 winner)	–	0.6187	–
Ours (Uncertainty-aware Residual Fusion)	**0.8489**	**0.6216**	**0.8366**

TABLE I
MAIN COMPARISON ON RUMOUREVAL-2019 TASK A.

Table I shows that our method substantially outperforms all in-house baselines. Compared with the strongest baseline (BERT fine-tuning), our model gains +0.1468 in Accuracy and +0.1745 in Macro-F1. Compared with the SemEval-2019 winning system (BLCU-NLP), our method achieves a slightly higher Macro-F1 (+0.0029 absolute).

C. Error Analysis

To further understand where errors still remain, we analyze the prediction file of our best model (`fusion=entropy_gate`, `gate_style=residual`, $\tau = 1.2$, $\gamma = 0.55$), and show the confusion matrix in Table II.

True \ Pred	**S**	**D**	**Q**	**C**
S	53	2	1	101
D	0	49	2	50
Q	5	1	54	33
C	11	30	40	1395

TABLE II
CONFUSION MATRIX OF THE BEST PROPOSED MODEL ON THE TEST SET. S=`SUPPORT`, D=`DENY`, Q=`QUERY`, C=`COMMENT`.

From Table II, we can observe three clear findings.

First, the main error is that many minority samples are predicted as `comment`. Specifically, 64.3% of true `support` (101/157), 49.5% of true `deny` (50/101), and 35.5% of true `query` (33/93) are classified into `comment`. This is the key reason why minority-class recall is still limited, even though the overall accuracy is high.

Second, confusion among minority classes themselves is not serious (e.g., only 2 `support`→`deny`, 2 `deny`→`query`, and 1 `support`→`query`). This means when the model already decides a sample is not `comment`, it can usually distinguish `support`/`deny`/`query` better.

Third, some true `comment` samples are over-predicted as stance classes (`comment`→`query`: 40; `comment`→`deny`: 30), which reduces the precision of minority classes.

In summary, the remaining challenge is mainly the decision boundary between weak stance and general commentary, rather than the separability among minority stance classes.

V. CONCLUSIONS

We proposed an uncertainty aware residual fusion framework for conversational stance detection. The model combines contextual semantic representations from RoBERTa with lexical signals derived from TF IDF, and dynamically adjusts lexical contribution according to semantic uncertainty estimated via entropy. The residual scaling design preserves a stable lexical pathway while enabling adaptive amplification when semantic confidence is low.

Experiments on RumourEval 2019 demonstrate that this simple yet principled fusion mechanism substantially improves minority class recognition under severe label imbalance and achieves competitive performance against strong baselines. These findings suggest that uncertainty guided feature interaction is an effective strategy for balancing complementary views in stance classification.

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